
**From Records to Reliable Decisions:
Measuring and Improving Port Data Maturity**

Summary

ICM Corporation needs to evaluate its data and analytics maturity, identify feasible improvements, establish monitoring protocols, and adapt the assessment to other ports and trucking firms. To address these tasks, we build four models: a coverage-constrained indicator selection model and three linked models for capability estimation, chain reliability, and robust resource allocation. Because the company withholds numerical records, the framework is demonstrated with explicitly labeled synthetic data.

First, we map port operations to nine information-management requirements and construct a dictionary of 16 indicators. The coverage model retains ten high-frequency indicators while reducing illustrative collection cost from 29 to 17 units. We then establish event identifiers, source records, missing-data bounds, and versioned processing rules. Deduplication yields 34,560 unique audit events, including 987 unknown outcomes that remain in the expected-event denominator.

Next, the Lag-aware Capability Filter (LCF) uses monthly audit counts to estimate people, technology, and process capability. The state equation combines persistence and delayed action effects, while the observation equation adjusts for workload and sample size. The resulting final capability profile is (0.4823, 0.5384, 0.4070), providing a common input for the maturity assessment rather than three unrelated KPI scores.

Then, the Chain Reliability Maturity (CRM) model links those capabilities to timely, correct import and export handoffs. It incorporates a people-process interaction and a shared-outage effect. Fixed reference conditions give a comparable maturity score of 41.96/100 in the demonstration, while separate operational scores reveal current workload and outage risks. The score is not presented as ICM's observed maturity.

Furthermore, the Robust Improvement Portfolio (RIP) model combines diminishing returns, implementation delays, dependencies, staffing, and quarterly budgets. Enumerating 16,807 menus selects training, governance, and recovery investments costing 5.45 units, with a worst planning-scenario net benefit of 5.9671 units; one unit represents USD 10,000. Nominal maturity rises to 55.32/100. We translate the decision process into a monitoring protocol and explain how compatible handoff measures support different port sizes and trucking partners.

Finally, model testing covers temporal prediction, probability calibration, parameter sensitivity, and held-out decision scenarios. Filtered capability RMSE is 0.0284, versus 0.0410 for direct monthly estimates; delayed forecast RMSE is 0.0349, versus 0.0470 without onset delay. The two chain Brier scores are 0.2289 and 0.2114. Tests also identify boundaries: nominal 90% state intervals cover only 66.7% of test states, and the robust menu violates budget in 5% of wider stress cases. These results support the framework's comparative use while requiring recalibration before enterprise deployment.

Keywords: indicator selection; state estimation; chain reliability; robust optimization; sensitivity analysis

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1 Introduction

1.1 Problem Background

A container's physical journey and its information journey must stay aligned. Arrival notices, manifests, customs releases, yard locations, and inland schedules are produced by different actors. A correct record arriving after a truck has departed can be as costly as an incorrect record. The ICM problem therefore asks for a system that evaluates people, technologies, processes, and their connection [1].

Our assessment covers two complete chains: import, from arrival information to inland release; and export, from inland receipt to vessel loading confirmation. A chain succeeds only when all required information is correct enough for the agreed task and available by its deadline. We do not equate analytics maturity with the number of dashboards, licenses, or sensors.

1.2 Problem Restatement

1. Define observable KPIs and an interpretable maturity metric.
2. Translate a measured profile into a feasible improvement portfolio.
3. Specify collection, governance, validation, and monitoring protocols.
4. Adapt the system to different port sizes and to trucking firms.
5. Explain these safeguards to customers in a one-page letter.

ICM supplies an operational description but withholds internal records. Consequently, its actual maturity is *not identifiable from the problem statement*. We deliver a measurement instrument and test its behavior using labeled synthetic records. No simulated interview, procurement quote, or corporate financial observation is presented as real evidence.

1.3 Our Work

Port digitalization involves several organizations whose records must agree at each handoff [2]. We begin with a dictionary of 16 indicators covering people, technology, processes, and service outcomes. The distinction between a quality dimension, its metric, and its measurement [3] gives each indicator a clear interpretation. A coverage-constrained selection model then reduces collection effort while retaining the required governance and recovery checks.

Monthly audit rates combine capability changes with sampling noise and workload effects. We use the Lag-aware Capability Filter (LCF), a state-space model based on Bayesian filtering [5], to estimate three capability trajectories and the delayed effects of recorded actions. The Chain Reliability Maturity (CRM) model maps these estimates to timely, correct import and export handoffs. Its probabilities provide a service-based maturity score, which is checked with proper scoring rules [6].

Improvement projects compete for the same budget and staff. The Robust Improvement Portfolio (RIP) model compares training, hiring, integration, governance, and recovery under cost and implementation uncertainty [7]. Its selected portfolio feeds into a collection and monitoring protocol. Provenance records [4] allow a reported score or decision to be traced back to the records used. We also discuss how the assessment can be adapted to different port sizes and trucking companies.

Figure 1 shows how indicator records feed the capability filter, how capabilities determine chain reliability, and how the investment decision returns to monitoring. The feedback arrow represents updating evidence and model parameters after implementation. Monitoring and cross-enterprise transfer remain proposals, not field-validated deployments.

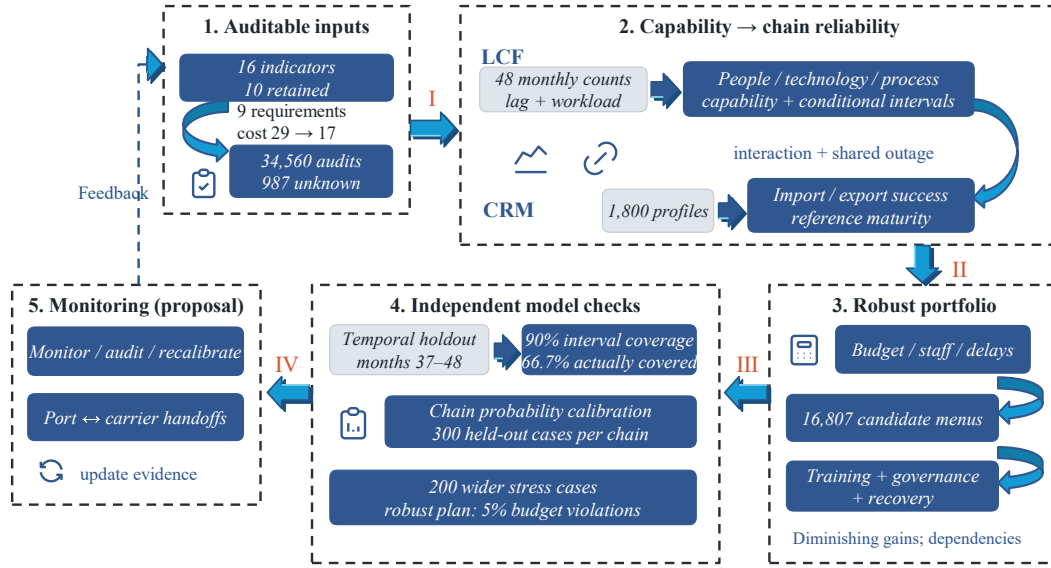


Figure 1: The evidence-to-decision framework

2 Assumptions and Notation

- A1. Comparable event definitions.** An import/export success has a versioned deadline and error rule. Otherwise probabilities across months are not comparable.
- A2. Anchored capabilities.** One audited pass probability fixes each latent dimension's direction and scale. The demonstration fixes its intercept to zero and loading to one.
- A3. Local response stability.** Within the four-quarter planning horizon, the action-response function is stable. Structural change triggers refitting.
- A4. Audits are conditionally independent in the generator.** Real repeated workers, shifts, or voyages require cluster effects; our reported intervals do not include them.
- A5. Loss is incremental and nonoverlapping.** Avoided losses apply to distinct import/export transactions. Shared-event losses must be allocated once.

Table 1: Core notation. All probabilities and capabilities are dimensionless.

Symbol	Meaning	Unit
$d \in \{P, T, R\}$	People, technology, process	—
t, q, k, j	Month, quarter, chain, project index	—
Y_{dt}, N_{dt}, m_{dt}	Passes, observed audits, unknown records	count
z_{dt}, a_{dt}	Logit capability; $a = \text{logit}^{-1}(z)$	—
p_k, M	Chain success probability; maturity score	—, 0–100
x_j, s_j, G_j	Project level, start quarter, gain ceiling	unit, quarter, —
B_q, Z_s	Quarterly budget; scenario net benefit	USD 10,000

3 Indicator Selection and Data Preparation

3.1 Selection of Maturity Indicators

A useful maturity assessment must cover several kinds of failure, but collecting every available KPI consumes audit effort. Indicators with similar monthly trends can still serve different purposes: uptime records ordinary availability, whereas a recovery test checks what happens after an outage. We select the high-frequency panel by requirement coverage and collection cost, keeping critical checks mandatory.

We describe each candidate by its operational definition, the requirements it covers, and the effort needed to

collect it. A binary variable records whether it enters the panel. The resulting selection is both a minimum-cost coverage problem and a list of measurements the port can actually maintain.

Each indicator must answer a business question, be observable, and support an action. A high correlation does not make recovery tests interchangeable with uptime. Table 2 defines the complete dictionary; a star marks the selected high-frequency panel. C2 is an outcome, not an additional independent capability measurement.

Table 2: Operational KPI definitions and management use.

ID	Operational definition	Action or source
P1*	Passed job tasks / valid practical tests	Targeted training; test log
P2*	Qualified staffed skill-slots / required slots	Hire, transfer, or contract
P3	Retained competence after training	Follow-up practical tests
P4	Approval-to-arrival staffing time	Recruitment records; censor unfinished cases
T1*	Available required-service time / required time	Service and outage logs
T2*	Correct delivered logical transactions / due transactions	Interface acceptance tests
T3	Actual minus required data-availability time	Median and tail latency
T4*	Recovery drills meeting requirements / valid drills	Recovery design and rehearsal
R1	Usable required fields / all required fields	Entry and schema rules
R2*	Independently verified correct items / audited items	Source-document audit
R3	Consistent matched records / required matches	Reconcile shared identifiers
R4*	Assets with accountable owners / applicable assets	Ownership register
R5*	Assets meeting metadata rules / checked assets	Catalog and schema version
R6*	Compliant access/lifecycle checks / applicable checks	Approval and disposal evidence
C1	Issue creation to independently confirmed closure	Escalation and root-cause review
C2*	Timely, correct complete chains / due chains	Customer handoff reliability

Let A_{gi} encode coverage of nine requirements: competence, staffing, reliability, interoperability, quality, accountability/security, metadata, lifecycle, and handoffs. With binary inclusion z_i , solve

$$\min_z \sum_i c_i z_i \quad \text{s.t.} \quad \sum_i A_{gi} z_i \geq 1 \quad (\forall g), \quad z_i = 1 \quad (i \in \mathcal{I}_{\text{mandatory}}). \quad (1)$$

Step 1: encode requirements Set $A_{gi} = 1$ exactly when indicator i supplies evidence for requirement g ; record effort c_i in a common unit. Lock the mandatory indicators before searching.

Step 2: enumerate and screen Enumerate all binary vectors $z \in \{0, 1\}^{16}$; discard a vector if any mandatory item is absent or any requirement has zero coverage.

Step 3: select and perturb costs Minimize total cost among surviving vectors, output their selected indicator IDs, then repeat after raising each individual cost by 20%. Compare the selected sets to assess local stability.

Table 2 marks the selected indicators with stars. Their definitions show which operational checks remain in the high-frequency panel.

Figure 2 makes the coverage structure explicit: retaining every indicator is unnecessary, but each requirement must retain an auditable measure. Icon tiles indicate coverage and blanks indicate none. Colors identify indicators, not performance; an asterisk marks selection and a plus marks mandatory retention.

The mandatory set is P2, T1, T4, R2, R4, R6, C2. Enumerating 2^{16} subsets selects ten indicators at 17 cost units versus 29 for all indicators, a 41.4% reduction. Raising each cost separately by 20% leaves the set unchanged (minimum Jaccard similarity 1.00). These costs are illustrative effort units. Low-frequency diagnostics remain available; deleting them from the high-frequency panel does not delete the underlying records.

Unchanged selection under individual cost perturbations indicates local stability. Joint cost changes can still alter the panel. The result also depends on how coverage is assigned: a checked requirement is meaningful only if the associated audit measures it adequately. Indicator definitions and audit quality should therefore be reviewed together.

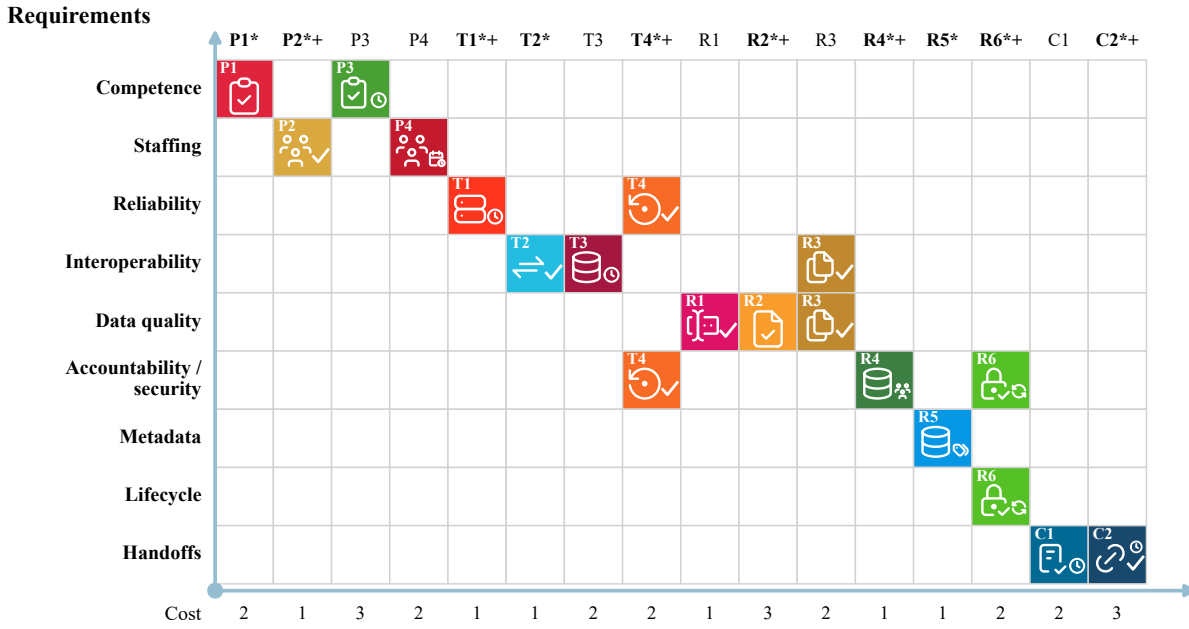


Figure 2: Coverage of requirements by candidate indicators

3.2 Data Collection

3.2.1 Event Records and Data Confidentiality

A delivery log contains only events that reach the system. A handoff that fails before a record is written can disappear from the denominator, producing an overstated success rate. We establish the expected-event registry before delivery and assign each handoff a business identifier, sender, recipient, required fields, and deadline. This registry records what should happen as well as what was observed.

Technical retries create a different counting problem: several attempts can correspond to one handoff. Each attempt receives its own event identifier but retains the shared business identifier. Counting handoffs by business identifier avoids inflating the transaction total, while the separate attempt records preserve evidence for diagnosing interface failures.

Late records can change a monthly result after it has been published. Each record stores the event and business identifiers, business unit b , event and ingestion times, source, outcome, quality flag, and rule version. Event time determines the reporting month; ingestion time identifies late arrivals. A closed report is retained, and subsequent corrections receive a new version. Unresolved outcomes remain marked as unknown.

Personnel tests and shipment records can disclose identities or commercial information even when the evaluator needs only counts and error categories. Personnel records use pseudonymous identifiers, with the identity mapping held separately under controlled access. Detailed calculations remain inside ICM; only approved aggregates leave the company. The internal audit trail retains enough detail to verify those aggregates without routinely exposing sensitive records.

3.2.2 Stratified Sampling

Daytime records and easily reached employees may dominate a convenient audit sample, leaving night shifts or complex cargo underrepresented. We collect logs exhaustively where practical and stratify manual audits by role, shift, terminal, cargo class, and workload. Population weights then correct differences between the sample composition and the operations being assessed. Let N_h be the stratum population and y_h/n_h its audited pass rate.

Step 1: calculate stratum weights Let N_h and n_h denote the population and sample size of stratum h , and y_h its observed passes. Calculate the population weight W_h and the sample pass rate $\hat{p}_h$:

$$W_h = \frac{N_h}{\sum_{g=1}^H N_g}, \quad \hat{p}_h = \frac{y_h}{n_h}. \quad (2)$$

Step 2: aggregate the estimate The weighted estimate uses population composition rather than the observed response mix:

$$\hat{p} = \sum_{h=1}^H W_h \hat{p}_h. \quad (3)$$

Step 3: quantify sampling variance For independent sampling within strata, let s_h^2 be the unbiased sample variance of the binary outcomes. The finite-population correction accounts for sampling without replacement:

$$s_h^2 = \frac{n_h}{n_h - 1} \hat{p}_h (1 - \hat{p}_h), \quad \widehat{\text{Var}}(\hat{p}) = \sum_{h=1}^H W_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h}, \quad n_h > 1. \quad (4)$$

With per-audit cost c_h and estimated within-stratum standard deviation S_h , continuous optimal allocation satisfies $n_h \propto N_h S_h / \sqrt{c_h}$, subject to a minimum sample in every relevant stratum. Repeated employees or voyages require a cluster bootstrap instead of the independent-record variance above. If a stratum has no responses, report an identification interval rather than a fabricated weighted estimate.

Table 3: Collection risks, corrections, and acceptance evidence.

Risk	Correction	Verification
Night shifts respond less often	Record response by stratum; supplement undercovered shifts	Compare sample and population shares
Confidential records cannot leave ICM	Compute inside ICM; release approved aggregates	Reproduce selected totals from internal evidence
Same auditor defines and validates rules	Freeze rules before independent re-audit	Keep validation sample separate

The demonstration uses one homogeneous audit stratum with 240 expected tests per dimension-month. Stratified reweighting is a deployment protocol, not an experiment claimed to have been completed. A governance owner approves purpose, retention, and access; the evaluation team sees only the information necessary for its task.

3.3 Data Cleaning and Missing Values

A cleaned total alone cannot explain whether a change came from operations, a merged duplicate, or a revised rule. We keep raw, cleaned, and analytical layers separate. The raw layer is append-only; the cleaned layer records corrections and unresolved conflicts; the analytical layer supplies counts, uncertainty, denominators, and rule versions. Provenance links these transformations to their source records [4].

1. **Meaning and time.** Different time zones or event codes can make the same handoff appear inconsistent. Standardize these definitions and units before comparing records. A negative delay is flagged for investigation because it may indicate a clock error or a genuine early arrival.
2. **Duplicate records.** Duplicate imports inflate the denominator, while conflicting copies leave the true result uncertain. Merge exact duplicates with the same event identifier, quarantine conflicts, and retain unresolved cross-system matches for review.
3. **Missing records.** Not-applicable items, lost logs, late reports, and open cases have different statistical meanings. Record the cause before forming denominators. Unfinished hiring requests and issue durations are right-censored rather than treated as completed cases.
4. **Extreme observations.** Outages and long customs-related delays are precisely the failures the assessment needs to detect. Outlier flags prompt source checks; confirmed events remain in the data.

Bounds for Missing Outcomes. Dropping unknown outcomes assumes that the observed records represent the missing ones, which is questionable when failures are harder to record. We instead calculate the two extreme cases, assigning all unknowns to failure or success. With N required items, y known passes, and m unknowns, the resulting bounds are

$$p \in \left[\frac{y}{N}, \frac{y + m}{N} \right]. \quad (5)$$

In simulated month 40, the process anchor’s observed-only rate is 45.96%, whereas the defensible full-denominator interval is [37.92%,55.42%]. Missing failures are deliberately more frequent in months 40–41. An apparently precise complete-case estimate conceals this uncertainty. We show the filter’s failure under that mechanism rather than silently imputing favorable outcomes.

Table 4: Record-conservation audit for the synthetic event file.

Stage	Records
Received records, including duplicates	35,597
Exact duplicates removed	1,037
Unique expected audit events	34,560
Observed outcomes	33,573
Unknown outcomes retained	987
Late flags retained (overlapping category)	920

The equations $35,597 - 1,037 = 34,560$ and $33,573 + 987 = 34,560$ are checked in the reproduction audit. Lateness is only flagged in this synthetic file; receipt-time forecasting is not tested. Monthly train/validation/test blocks are 1–24, 25–36, and 37–48. Parameters and delay choices never use test outcomes.

4 Dynamic Estimation and Forecasting of Port Data-Management Capability

4.1 Measurement of People, Technology, and Process Capability

This section estimates the current level of three port data-management capabilities and forecasts their next-period values from monthly audit records. People capability concerns staff’s practical data-handling skills, technology capability concerns reliable information delivery, and process capability concerns the execution of quality and governance rules. A defined audit pass probability anchors each dimension; other indicators provide operational checks. The estimates represent underlying performance at the reference workload rather than one month’s observed pass rate.

The estimated profile identifies dimensions needing attention. Section 5 converts it into handoff reliability and maturity; Section 6 uses it as the baseline for improvement-project optimization. Next-period forecasts support monitoring. Recorded actions help describe delayed changes, while their causal effects remain unverified.

Monthly pass rates fluctuate with audit size and workload, while training and system changes take time to affect operations. A static score confounds these sources of variation. Inspired by state-space filtering, LCF separates an evolving latent capability from noisy measurements and places action delay in the transition equation. The objective is to recover a useful capability trajectory and forecast it without treating short-run KPI changes as immediate intervention effects.

4.2 State-space Model

The observations are monthly passes and audit denominators, accompanied by workload and dated actions. A capability estimate should remain on the probability scale, although the state transition needs an unrestricted variable. We express capability as a latent logit state and transform it back after filtering. The outputs are three capability estimates, one-step forecasts, and conditional intervals for use in CRM.

The logistic link maps an unrestricted latent state to a capability between zero and one:

$$a_{dt} = \sigma(z_{dt}) = \frac{1}{1 + \exp(-z_{dt})}. \quad (6)$$

In the transition below, ϕ_d is persistence, μ_d is the baseline logit, β_d is action strength, and Q is state-disturbance variance. u_{dt} records an action, and δ_d gives its onset delay.

An action can start influencing operations after its implementation date, and its effect can persist across several months. We distinguish the onset delay from the subsequent three-month input window:

Step 1: form the delayed input Use the onset delay and the normalized weights (1, 0.6, 0.36) to combine three action months:

$$\tilde{u}_{dt} = \frac{\sum_{\ell=0}^2 0.6^\ell u_{d,t-\delta_d-\ell}}{1 + 0.6 + 0.6^2}, \quad u_{dt} = 0 \quad (t < 1). \quad (7)$$

Step 2: evolve capability The previous state, baseline reversion, and delayed action determine the next state:

$$z_{dt} = \phi_d z_{d,t-1} + (1 - \phi_d) \mu_d + \beta_d \tilde{u}_{dt} + \eta_{dt}, \quad \eta_{dt} \sim N(0, Q). \quad (8)$$

The onset delay δ_d shifts the action input, while the three-month weights describe its subsequent contribution. Persistence carries earlier effects forward through the state equation. Monthly pass counts also depend on workload: the same capability can yield fewer passes under peak demand. We represent that distinction with the observation equation

$$Y_{dt} | z_{dt} \sim \text{Binomial}(N_{dt}, \sigma(z_{dt} - 0.25c_t)), \quad c_t = \sin(2\pi(t-1)/12). \quad (9)$$

The workload coefficient -0.25 and process-noise standard deviation $\sqrt{Q} = 0.08$ are declared scenario assumptions. They are not estimates from ICM. In deployment, estimate them using a pilot and audit-based workload contrasts.

4.3 Capability Estimation by Filtering

The binomial observations and latent logit state have different scales. An empirical-logit transformation gives a Gaussian working approximation in which prediction and observation can be combined analytically. This makes sequential updating practical for the moderate audit counts used here. The following steps are applied to each dimension; its subscript is omitted for readability.

4.3.1 Step 1: construct the observation and its variance

Given passes Y_t , observed trials N_t , and workload c_t , add a half-count correction to avoid infinite logits. r_t is the workload-adjusted observation, and R_t is its working variance:

$$r_t = \log \frac{Y_t + 1/2}{N_t - Y_t + 1/2} + 0.25c_t, \quad R_t = \frac{1}{Y_t + 1/2} + \frac{1}{N_t - Y_t + 1/2}. \quad (10)$$

The approximation is $r_t \approx z_t + \epsilon_t$, with $\epsilon_t \sim N(0, R_t)$. Small or extreme counts require a binomial filter in a later deployment.

4.3.2 Step 2: predict the state

Given the preceding mean m_{t-1} and variance V_{t-1} , apply the fitted persistence, baseline, action strength, and disturbance variance:

$$m_t^- = \phi m_{t-1} + (1 - \phi) \mu + \beta \tilde{u}_t, \quad V_t^- = \phi^2 V_{t-1} + Q. \quad (11)$$

The implementation initializes $m_0 = \mu$, $V_0 = 0.5$. These predictions use only information available before the current observation.

4.3.3 Step 3: update the state and report capability

A month with few audits should not outweigh a precise state prediction. Innovation variance S_t combines prediction and observation uncertainty, and the Kalman gain K_t determines the weight assigned to the new observation:

$$S_t = V_t^- + R_t, \quad K_t = \frac{V_t^-}{S_t}. \quad (12)$$

Then combine the prediction and innovation $r_t - m_t^-$ to obtain posterior mean and variance:

$$m_t = m_t^- + K_t(r_t - m_t^-), \quad V_t = (1 - K_t)V_t^-. \quad (13)$$

Finally convert the logit state into capability and a nominal 90% conditional interval:

$$\hat{a}_t = \frac{1}{1 + e^{-m_t}}, \quad I_t^{90} = \left[\frac{1}{1 + e^{-(m_t - 1.645\sqrt{V_t})}}, \frac{1}{1 + e^{-(m_t + 1.645\sqrt{V_t})}} \right]. \quad (14)$$

A smaller raw residual does not necessarily imply a better model when observation precision varies by month. The innovation loss weights residuals by their variance and penalizes excessive uncertainty. For each candidate delay, fit $\theta = (\phi, \mu, \beta)$ on the training months, then select the delay using the same loss on the validation months:

$$\hat{\theta}_\delta = \arg \min_{\theta} \frac{1}{2} \sum_{t=1}^{24} \left[\log S_t + \frac{(r_t - m_t^-)^2}{S_t} \right], \quad 0.5 \leq \phi \leq 0.98, \quad -2 \leq \mu \leq 2, \quad 0 \leq \beta \leq 0.6. \quad (15)$$

Algorithm 1: Estimate and forecast capabilities

Input: monthly counts, workload, dated actions. Fit (ϕ, μ, β) on months 1–24 by minimizing the Gaussian innovation negative log likelihood for each $\delta \in \{0, 1, 2, 3\}$. Select delay on months 25–36. Freeze parameters, then filter and predict months 37–48 sequentially. **Output:** $\sigma(m_t)$, one-step predictions, and conditional intervals $\sigma(m_t \pm 1.645\sqrt{V_t})$.

4.4 Estimation Results and Model Comparison

For test states indexed by $\mathcal{T}$, RMSE measures average capability error and coverage measures how often the known generating state falls within the reported interval:

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(d,t) \in \mathcal{T}} (\hat{a}_{dt} - a_{dt}^{\text{true}})^2}, \quad \text{Coverage} = \frac{1}{|\mathcal{T}|} \sum_{(d,t) \in \mathcal{T}} \mathbf{1}\{a_{dt}^{\text{true}} \in I_{dt}^{90}\}. \quad (16)$$

The following numerical experiments use synthetic data.

Figure 3 compares recovered capabilities with the known synthetic trajectories. Its three panels represent people, technology, and process; the shaded months form the common test set for the direct monthly, zero-delay, and delayed models compared below. Bands are nominal 90% intervals conditional on fitted parameters. Green lines separate training (1–24), validation (25–36), and shaded test months (37–48); all panels share the vertical range 0.20–0.80.

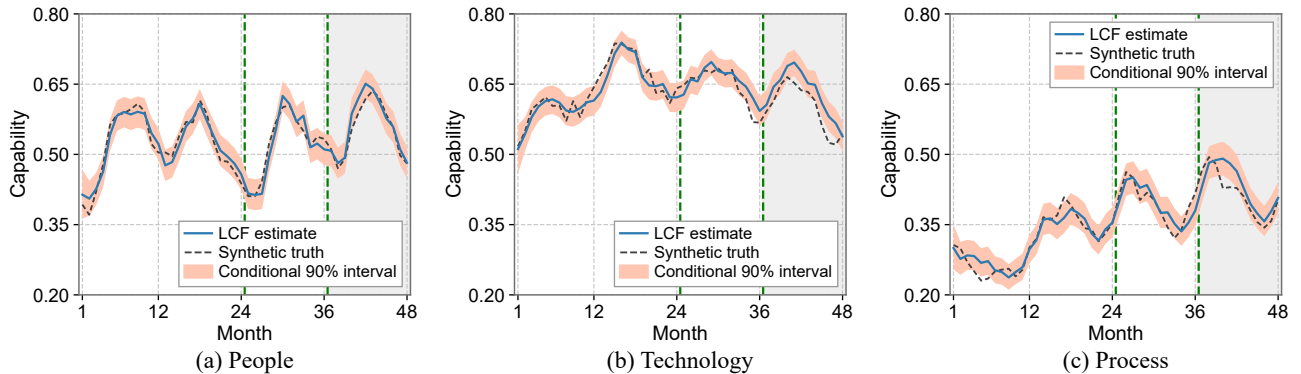


Figure 3: Recovery of three-dimensional capability over 48 months

Table 5: Test-block errors and delay recovery.

Quantity	Result
Filtered capability RMSE / raw monthly RMSE	0.0284 / 0.0410
Delayed one-step RMSE / zero-delay one-step RMSE	0.0349 / 0.0470
Coverage of nominal 90% conditional intervals	66.7%
Selected delays (P,T,R); generating delays	(2, 3, 2); (2, 1, 2)

The lower pooled errors support using the filtered profile for subsequent scoring: the state estimate suppresses audit noise, and the delayed forecast better reflects the timing of actions. Figure 3 also preserves dimension-specific trajectories, so improvement in the aggregate does not hide a weak process capability.

4.5 Forecast Errors and Interval Coverage

Filtering reduces sampling noise, and validation-selected onset delays improve the pooled forecast error. Yet technology's selected delay is three months instead of the generating one month. A short, periodic action series does not reliably identify a delay: persistence, baseline, and action strength can compensate for each other. We therefore interpret the model as a forecasting instrument, not proof of a training or software intervention's causal effect.

Only 24 of 36 test states lie inside nominal 90% conditional intervals. Failure-dependent missingness, parameter uncertainty, and the Gaussian approximation all contribute plausible explanations; this experiment does not isolate their shares. Before operational scoring, use independent failure audits, full-denominator bounds, and a validation-calibrated predictive interval. Until coverage is acceptable on fresh data, publish the point estimate with an uncertainty warning rather than a confident certification.

Figure 4 distinguishes state-estimation error from uncertainty due to missing audit outcomes. Each panel contains twelve test months; the lower row uses the full expected denominator, not an imputed success rate. The upper row compares filtered errors and conditional 90% intervals with raw monthly errors; hollow diamonds mark missed truth. The lower row compares observed-only rates and full-denominator missing bounds with known generating probabilities, not recovered complete-data rates. These bounds are not confidence intervals. Gray bands mark enhanced failure-dependent missingness in months 40–41, without causal attribution. Panel vertical ranges differ.

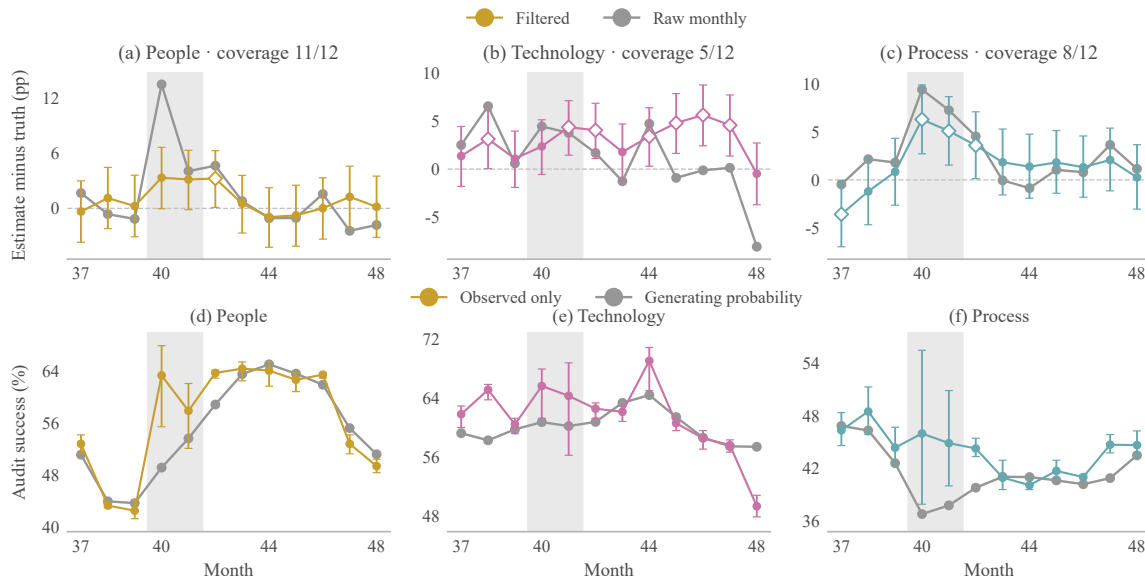


Figure 4: State-estimation errors and missing-outcome bounds

5 Maturity Assessment Based on Chain Reliability

5.1 Chain Reliability as a Maturity Measure

A capability profile describes resources inside the port, but customers experience the success or failure of an entire handoff. We define maturity through the probability that the required information is correct and available by the deadline. This connects the assessment to an outcome that can be independently audited. A weighted capability sum does not specify that success probability. Multiplying stage success rates introduces an independence assumption that shared outages can violate. CRM models the whole-chain outcome directly; a logistic response keeps predictions in $[0, 1]$, while interactions and environmental terms account for complementary capabilities, workload, and shared failures.

5.2 Reliability Model and Maturity Score

A trained employee still needs an assigned data owner and a usable correction process to resolve an exception. The people–process interaction represents this complementarity. Workload and outage terms distinguish changes in operating conditions from changes in capability. For the capability vector a , workload c , and outage flag o , we define each chain probability by

$$\text{logit } p_k(a, c, o) = b_{0k} + b_{Pk}a_P + b_{Tk}a_T + b_{Rk}a_R + h_k a_P a_R + \gamma_k c + \psi_k o. \quad (17)$$

Here b_{0k} is the chain intercept, b_{Pk}, b_{Tk}, b_{Rk} are capability coefficients, h_k is the interaction coefficient, and γ_k, ψ_k measure workload and outage effects. With linear predictor η_k equal to the right-hand side, the probability is $p_k = 1/(1 + \exp(-\eta_k))$.

An unconstrained fit could associate stronger skills or governance with lower success because of finite-sample noise. Within the declared model domain, we require nonnegative capability and interaction coefficients and nonpositive workload and outage effects: $b_{Pk}, b_{Tk}, b_{Rk}, h_k \geq 0$, $\gamma_k, \psi_k \leq 0$. The resulting marginal effects include

$$\frac{\partial p_k}{\partial a_P} = p_k(1 - p_k)(b_{Pk} + h_k a_R) \geq 0, \quad \frac{\partial p_k}{\partial a_R} = p_k(1 - p_k)(b_{Rk} + h_k a_P) \geq 0. \quad (18)$$

These inequalities prevent an estimated association from recommending deliberately worse skills or governance within the model domain. They are structural restrictions, not empirical proof that every real intervention helps.

Reference and operational scores. Ports assessed during peak demand or an outage should not be compared directly with those assessed under normal conditions. We calculate a reference score at fixed workload and outage settings, alongside an operational score using the current settings:

$$M^{\text{ref}}(a) = 100 \sum_{k=1}^2 \omega_k p_k(a, 0, 0), \quad M_t^{\text{op}} = 100 \sum_{k=1}^2 \omega_k p_k(a_t, c_t, o_t), \quad \omega = (0.6, 0.4). \quad (19)$$

Reference maturity fixes workload and outage conditions. Operational performance retains the actual environment and must be reported alongside it. The demonstration weights follow an assumed import/export transaction mix, not loss severity; monetary loss weights appear separately in optimization. During a real outage, a stable reference score must never conceal an operational red alert.

We report a continuous score and three-dimensional capability profile. No supported level boundaries are supplied by ICM, so we do not invent five maturity grades. A future certified level would require both an empirically calibrated score threshold and mandatory security, recovery, ownership, and lifecycle checks. A high average cannot offset a failed critical gate. NIST’s historical 2018 framework informs the risk-management structure, not a numeric maturity conversion [8].

5.3 Parameter Estimation and Score Calculation

Step 1: construct predictors and split samples The interaction and environment terms must enter the fitted design matrix as well as the stated equation. For profile i , construct $X_i = (1, a_{Pi}, a_{Ti}, a_{Ri}, a_{Pi}a_{Ri}, c_i, o_i)$. Separate 1,200 training profiles, 300 validation profiles, and 300 test profiles before fitting, so that parameter selection does not use test outcomes. Chain k has binary response y_{ik} .

Step 2: fit the constrained response Capability variables and their interaction can share predictive information, making coefficients unstable. Ridge regularization controls coefficient size, with its strength chosen from validation performance. For each chain and $\lambda \in \{0, 0.001, 0.01\}$, L-BFGS-B minimizes the penalized training log loss below. The seven coefficients follow the order of X_i ; the intercept is unpenalized:

$$\hat{\theta}_{k,\lambda} = \arg \min_{\theta_k \in \mathcal{B}} \left\{ \frac{1}{1200} \sum_{i \in \mathcal{D}_{\text{train}}} [\log(1 + e^{X_i \theta_k}) - y_{ik} X_i \theta_k] + \frac{\lambda}{2} \sum_{j=1}^6 \theta_{kj}^2 \right\}. \quad (20)$$

The numerical domain is $\mathcal{B} = [-8, 3] \times [0, 6]^4 \times [-4, 0] \times [-6, 0]$. These bounds implement the sign restrictions and stabilize estimation in this demonstration.

Step 3: select, freeze, and evaluate Choose the penalty with minimum validation Brier score for each chain. Freeze its coefficients, predict the test outcomes, and compare with the training-frequency baseline using the metrics below. Apply the same frozen mapping to the LCF profile to calculate reference and operational scores.

Step 4: propagate uncertainty A single maturity score conceals uncertainty inherited from both state estimation and response fitting. Generate 1,000 joint state/coefficient draws and calculate a score for each. Report the 5th and 95th percentiles, then forward two whole draws near those percentiles to RIP as additional planning cases.

5.4 Reliability Estimates and Calibration

Correct classification alone cannot tell whether a predicted 90% success rate is credible. We assess the probabilities with Brier score and log loss, which penalize inaccurate predictions and overconfidence differently [6]. For n held-out handoffs, p_i is predicted success and $y_i \in \{0, 1\}$ is the observed outcome; lower values are better:

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2, \quad (21)$$

$$\text{LogLoss} = -\frac{1}{n} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \quad (22)$$

The following numerical experiments use synthetic data.

The pilot generator draws 1,800 independent capability profiles uniformly over $[0.2, 0.9]^3$, workload over $[-1, 1]$, and an outage indicator with probability 0.15. It generates two Bernoulli outcomes per profile from specified coefficients. Broad profiles are needed because one port's short history cannot identify all interactions. These profiles are experimental test cases, not a survey of 1,800 ports.

The constant-frequency baseline and CRM are evaluated on the same 300 test profiles. Calibration summaries contain only four import bins and five export bins; unequal bin counts limit their precision. We retain the numerical scoring results below rather than a sparse standalone calibration plot.

Table 6: Independent synthetic test performance ($n = 300$ per chain).

Chain	Brier	Constant baseline	Log loss	True- p RMSE
Import	0.2289	0.2506	0.6480	0.0742
Export	0.2114	0.2549	0.6117	0.0294

Brier score is $n^{-1} \sum_i (p_i - y_i)^2$; the baseline uses the training success frequency. Both fitted chains improve on that baseline. The import model's stronger regularization shrinks coefficients and gives a larger true-probability error. Calibration bins are descriptive, with unequal sample sizes; they do not establish cross-port validity. Proper scoring rules discourage confident but inaccurate probabilities [6].

5.5 Uncertainty in Maturity Scores

The final filtered profile is $(0.4823, 0.5384, 0.4070)$, giving reference maturity 41.96. We propagate 1,000 draws of the final Gaussian state and coefficient vectors based on the observed-information approximation. Within-coefficient covariances are retained; signs are projected onto the permitted domain. The resulting 5th–95th percentile range is $[39.50, 52.71]$. It is an approximate uncertainty envelope, not a calibrated 90% confidence interval. State-parameter uncertainty, shrinkage bias, and cross-chain coefficient dependence remain omitted.

Two whole joint draws nearest the 5th and 95th score percentiles are forwarded to portfolio stress testing. Selecting a few representative draws is tractable, but does not guarantee coverage of every adverse parameter combination.

6 Robust Optimization of Improvement Projects

6.1 Investment Priorities under Limited Resources

A maturity deficit does not determine which project should be funded first. Equal spending overlooks unequal response curves; nominal optimization can exhaust a budget when costs rise. RIP connects the CRM service

probabilities to incremental avoided loss and chooses a feasible menu against a declared scenario set. Diminishing returns discourage overconcentration, delays reward timely delivery, and dependencies prevent investment in integration without governance support.

6.2 Portfolio Optimization Model

Project decisions need to specify both the amount invested and when work begins. Using the current capability state and fitted chain probabilities, we calculate each project combination's service improvement, avoided loss, cost, and staff demand. Budgets and project dependencies restrict feasible combinations; the smallest net benefit across planning scenarios determines their ranking.

6.2.1 Project Effects and Implementation Delays

Training, hiring, and integration cannot deliver their full benefit on the start date. The response model separates the committed investment from the portion effective in quarter q . Project j has level $x_j \in \{0, 1, 2, 3\}$ and start quarter $s_j \in \{1, 2\}$ when active.

Step 1: calculate effective investment Let d_j be the project-specific delay and Δ_s the extra scenario delay. $f_{jq}^{(s)}$ is the realized fraction of an investment by quarter q , rising over two quarters; $E_{jq}^{(s)}$ is its effective amount:

$$f_{jq}^{(s)} = \min \left\{ 1, \max \left(0, \frac{q - s_j - d_j - \Delta_s + 1}{2} \right) \right\}, \quad E_{jq}^{(s)} = x_j f_{jq}^{(s)}. \quad (23)$$

Step 2: calculate diminishing capability gains The first increment can address a large unresolved gap, while later increments face fewer remaining opportunities. A saturating exponential represents this diminishing return. With gain ceiling $G_j^{(s)}$ and conversion rate k_j , the response and derivatives are

$$g_j^{(s)}(E) = G_j^{(s)}(1 - e^{-k_j E}), \quad (24)$$

$$\frac{dg_j^{(s)}}{dE} = G_j^{(s)} k_j e^{-k_j E} > 0, \quad \frac{d^2 g_j^{(s)}}{dE^2} = -G_j^{(s)} k_j^2 e^{-k_j E} < 0. \quad (25)$$

Thus additional investment increases capability, but each increment contributes less than the previous one.

Step 3: combine project contributions Let m_{jd} be project j 's allocation to capability d , and $a_d^{0,(s)}$ the scenario baseline. Add realized contributions once and cap the probability-scale state:

$$a_{dq}^{(s)}(x) = \min \left\{ 0.995, \max \left[0, a_d^{0,(s)} + \sum_j m_{jd} g_j^{(s)}(E_{jq}^{(s)}) \right] \right\}. \quad (26)$$

The capability increment is measured against the same baseline a^0 at every quarter. Adding the total cumulative gain repeatedly to the preceding quarter would create artificial growth. An equivalent recursive implementation must add only $g_j(E_{jq}) - g_j(E_{j,q-1})$.

6.2.2 Objective Function and Constraints

Step 1: calculate quarterly project costs For scenario cost multiplier κ_s , investment x_j , and activation $y_j = \mathbf{1}(x_j > 0)$, the cost includes a fixed 0.15-unit startup charge:

$$C_q^{(s)}(x) = \sum_{j:s_j=q} \kappa_s (x_j + 0.15 y_j). \quad (27)$$

Step 2: calculate discounted net benefit A higher maturity score has no direct monetary unit. We first convert the increase in chain success into avoided failures using transaction volume and failure loss, then discount future benefits and subtract project costs:

$$Z_s(x) = \sum_{q=1}^4 \frac{\sum_k V_{kq} L_{kq}^{(s)} [p_k(a_q^{(s)}(x), 0, 0) - p_k(a^{0,(s)}, 0, 0)]}{(1+r)^{q-1}} - \sum_{q=1}^2 \frac{C_q^{(s)}(x)}{(1+r)^{q-1}}. \quad (28)$$

Here V is the quarterly count and L the loss per failed handoff; their product is converted to USD 10,000 units. Safety and legal obligations are not traded against this monetary benefit. With $y_j = \mathbf{1}(x_j > 0)$, solve

Step 3: optimize the worst-case feasible return Selecting only the largest nominal benefit leaves a portfolio exposed to cost overruns and weaker effects. We introduce v as the lower bound on net benefit across all declared scenarios and maximize it. Quarterly budget B_q , staff capacity H_q , unit staff demand h_j , and project dependencies enter the constraints:

$$\max_{x,s,v} v, \quad v \leq Z_s(x) \quad (\forall s \in \mathcal{S}), \quad (29)$$

$$\sum_{j:s_j=q} \kappa_s(x_j + 0.15y_j) \leq B_q, \quad \sum_{j:s_j=q} h_j x_j \leq H_q, \quad (30)$$

$$y_{\text{integration}} \leq y_{\text{governance}}, \quad s_{\text{governance}} \leq s_{\text{integration}} \quad \text{if integration active}, \quad (31)$$

$$x_{\text{recovery}} \geq 1. \quad (32)$$

The first line maximizes the minimum scenario return. The second enforces quarterly budget and staffing capacity. The third requires a governance foundation before or alongside integration. The final line guarantees a minimum recovery project.

A recovery floor is an explicit scenario management decision, not a universal standard. Governance produces an initial usable foundation in its start quarter, so simultaneous integration is allowed in the demonstration. If real acceptance must precede integration, replace this with a strict commissioning dependency. The staffing constraint is a capacity proxy for workload and disruption; no separate downtime guarantee is claimed.

6.2.3 Parameter and Scenario Settings

The following numerical experiments use synthetic data.

Table 7: Declared project-response assumptions. Costs are USD 10,000 units.

Project	G_j	k_j	d_j	h_j	$m_j = (P, T, R)$
Training	.16	.80	1	1.2	(1, 0, 0)
Hiring	.18	.55	2	.7	(1, 0, 0)
Integration	.22	.60	1	1.5	(0, 1, 0)
Governance	.24	.75	0	1.0	(0, 0, 1)
Recovery	.12	.65	1	.8	(0, .65, .35)

Budgets are (7, 4) units in the first two quarters; staff capacities are (8, 6) units. The fixed start cost is 0.15 per active project, and the quarterly discount rate is 2%. Each quarter has 6,000 import and 4,000 export handoffs with failure losses of USD 40 and USD 60. These are test assumptions, not quotations or industry benchmarks. The model can leave funds unused when additional projects reduce worst-case net benefit.

Table 8: Planning scenarios. Joint uncertainty adds two complete sampled state/coefficient configurations.

Scenario	Cost	Gain	Extra delay	Value
Nominal	1.00	1.00	0	1.00
Overrun	1.20	.90	0	1.00
Delay	1.10	.80	1	1.00
Lower avoided loss	1.05	.85	0	.80
Joint draw, low / high score	1.10	.85	0	.90

6.3 Enumeration of Feasible Portfolios

The menu contains only five projects and a few levels and start dates. Exhaustive enumeration is therefore tractable and verifies the best feasible choice without a heuristic stopping rule. The calculation proceeds as follows:

Step 1: enumerate candidate menus For each of five projects, enumerate seven options: off, or investment level 1, 2, or 3 starting in quarter 1 or 2. Their Cartesian product gives $7^5 = 16,807$ menus.

Step 2: screen feasible menus Compute quarterly spending and staff use, then check every budget scenario, governance dependency, and the recovery floor. The 20% overrun envelope is the largest planning cost multiplier; 1,418 menus remain feasible.

Step 3: evaluate delayed scenario returns For each feasible menu and scenario, evaluate effective investment, capability gain, chain success, avoided loss, and discounted net benefit in that order using the equations above. Record the minimum scenario return for each menu.

Step 4: select and independently stress-test Select the menu with the greatest minimum return and output its levels, start dates, cost, and staffing. Exhaustive evaluation gives zero optimality gap within this finite menu. Freeze the selected menu before the 200 wider held-out shocks; record feasibility separately from return and regret.

6.4 Optimal Portfolio and Strategy Comparison

Table 9: Selected project menu; inactive projects have no start date.

Project	Units	Start quarter	Cost
Training	1	1	1.15
Hiring	0	–	0.00
Integration	0	–	0.00
Governance	3	1	3.15
Recovery	1	1	1.15

The selected portfolio spends 5.45 units (USD 54,500) in quarter 1, requiring five capacity units. A 20% overrun gives 6.54 cost units, below the seven-unit budget. The nominal final maturity is 55.32, a 13.36-point increase. This menu favors governance, training, and required recovery under the assumed horizon and returns; zero hiring or integration is not a recommendation to eliminate those functions.

6.4.1 Returns under Planning Scenarios

Figure 5 locates the robust and nominal optima within the feasible menu space. Nominal optimization achieves a higher nominal return but sacrifices worst-scenario return and fails the full planning constraints. The plot shows 2,133 nominal-feasible menus: 1,418 robust-feasible and 715 nominal-only feasible; 14,674 nominal-infeasible menus are excluded. Returns are in synthetic USD 10,000 units, and dashed projections locate the two optima.

Equal allocation gives each project one unit in quarter 1. Greedy search starts from mandatory recovery and adds the feasible one-level increment with greatest nominal gain, also in quarter 1. Nominal optimization uses nominal budgets and returns; robust optimization uses every planning scenario and the 20% cost envelope.

Table 10: Finite-menu comparison in USD 10,000 units.

Plan	Nominal net	Worst net	Q1 spend
Robust	16.3697	5.9671	5.45
Nominal	18.5126	4.0743	6.45
Equal	14.1157	3.5331	5.75
Greedy	16.5363	5.6919	5.60

6.4.2 Portfolio Performance under Joint Shocks

A portfolio can appear robust if it is tested only on the scenarios that determined its selection. We evaluate the frozen decisions on 200 wider held-out shocks. Severity $u \sim U(0, 1)$ changes cost to $1 + 0.3u$ and effects to $1 - 0.3u$; extra delay has probability 0.3, and value is uniform on $[0.75, 1.15]$. Half the cases use $G \min(kE, 0.85)$ instead of the exponential response, testing sensitivity to the assumed curve as well as its parameters.

For a feasible decision x in scenario s , regret is the difference from the best menu that is feasible in that same

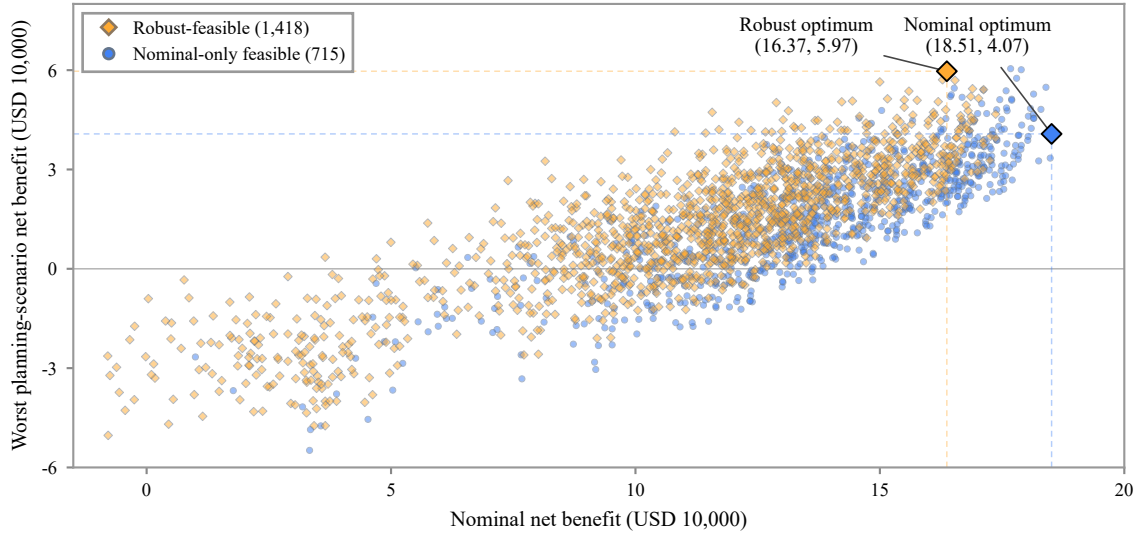


Figure 5: Nominal–worst-case return trade-offs across feasible portfolios

scenario:

$$\text{Regret}_s(x) = \max_{x' \in \mathcal{F}_s} Z_s(x') - Z_s(x), \quad x \in \mathcal{F}_s. \quad (33)$$

Here $\mathcal{F}_s$ is the scenario-specific feasible set. Infeasible decisions are counted as violations rather than assigned regret.

Table 11: Wider held-out scenarios: budget reliability and feasible regret.

Plan	Mean net	Violation rate	Mean feasible regret
Equal	9.542	32.5%	2.146
Greedy	11.001	19.0%	0.478
Nominal	9.760	69.0%	1.312
Robust	10.126	5.0%	1.253

Figure 6 compares returns draw by draw under the same shocks. Gold diamonds identify cases where only the robust plan remains feasible; feasibility and return are distinct outcomes. Both panels use the same 200 simulated shocks, not independent samples. Blue lines denote equal return rather than fitted regressions; shapes distinguish feasibility. Infeasible returns exclude financing and emergency penalties.

6.5 Budget Risk and Robustness

The held-out comparison in Figure 6 tests a wider cost envelope and an alternative response shape, rather than repeating only the scenarios used to optimize. The budget reoptimization results reported in the sensitivity section show whether a spending change alters the actual selected menu. Together these checks assess decision stability and feasibility beyond the nominal score gain.

Robust planning reduces violations from 69% to 5%, but greedy search earns more on average with greater budget risk. Regret compares only feasible cases with each scenario's exact optimum; infeasibility is reported separately. Mean return excludes financing and emergency penalties. The 5% rate is a stress-test observation, not a chance-constraint guarantee. Procurement decisions need contingency reserves or a wider cost envelope.

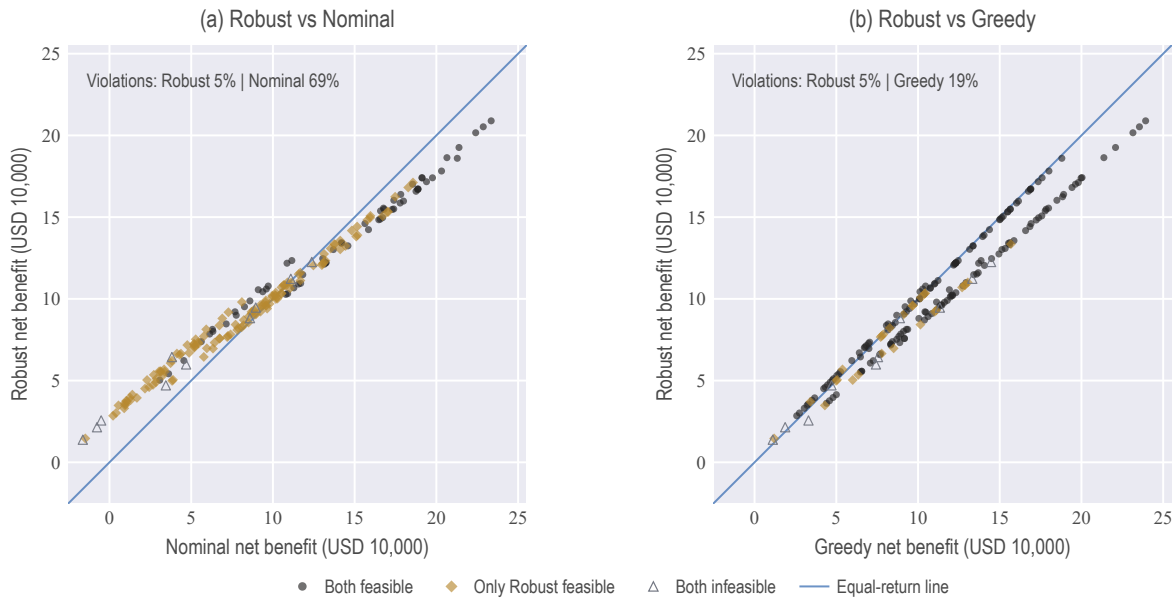


Figure 6: Paired portfolio returns under held-out shocks

7 Model Validation and Sensitivity Analysis

7.1 Validation under Different Operating Conditions

Average test errors do not reveal how the assessment responds to a weak capability or a service disruption. We vary capability balance, workload, outages, investment, and missingness in separate scenarios, then check whether the score changes in a way consistent with its definition.

Figure 7 shows the fitted CRM response to people and process capability with technology fixed at the estimated final capability, $a_T = 0.5384$, and workload fixed at zero. A shared outage lowers operational reliability over the same capability grid. This slice differs from the balanced $(.7, .7, .7)$ profile used in the scenario table below. Each panel contains 100 frozen-model predictions, not new observations, on the same 0–100 scale. White contours mark score levels; the final profile scores 41.96 without an outage and 19.18 with one.

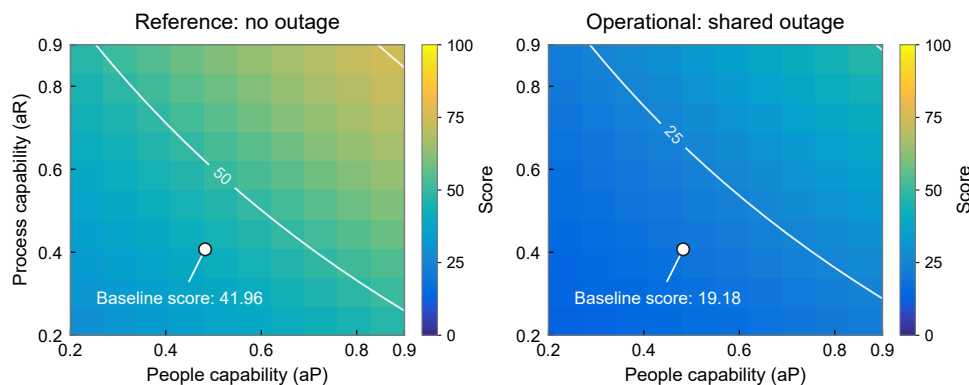


Figure 7: Capability–reliability response under shared outages

The following scenario contrasts are designed checks, not standardized rankings of different ports. Diminishing investment gains are already specified by the response function and summarized numerically.

The shared-outage test is especially useful: reference maturity isolates capability for comparison, but service reliability can collapse immediately. A customer dashboard should therefore display current incident status be-

Table 12: Scenario questions and observed model behavior.

Scenario and test	Result and implication
1 Balanced (.7, .7, .7)	65.32 reference points; comparison anchor
2 Strong technology, weak process (.7, .9, .3)	52.82; better hardware does not fully offset governance gaps
3 Training not yet effective (.45, .7, .7)	55.15; separately validate delayed forecasts
4 Peak workload $c = 1$	Operational score 53.69; reference score stays 65.32
5 Shared outage $o = 1$	Operational score 36.45; both chains deteriorate
6 Increasing training investment	First-unit gain .0881, third-unit gain .0178
7 Failure-selective missing logs	Month 40 process interval [37.92%,55.42%]

fore the reference score. The weak-governance comparison also explains why our portfolio does not allocate all resources to technology even when technical products are easier to purchase.

These are designed contrasts. They test whether the model responds in the intended direction, not whether the corresponding score levels are realistic for a specific port. The model would require external outcome calibration before those values support a contract or public assurance claim.

7.2 Sensitivity to Budget and State Noise

A decision is more sensitive than a smooth score when discrete project levels cross budget boundaries. At 80% or 90% of the original budgets, the robust portfolio changes from (1, 0, 0, 3, 1) to (1, 0, 0, 2, 1), and worst net benefit falls to 5.6443 units. A 10% budget increase does not change the menu; a 20% increase adds one training unit and raises worst net benefit to 6.0447 units. Budget expansion therefore has a discontinuous, diminishing decision value.

Table 13: Sensitivity to the assumed state-noise standard deviation; other fitted parameters are frozen.

Change	$\sqrt{Q}$	Test RMSE	Final reference score
-20%	.064	.0290	42.38
0%	.080	.0284	41.96
+20%	.096	.0285	41.63

Small score changes under this one perturbation do not repair the failed interval-coverage test. Sensitivity and calibration answer different questions. The optimization's exponential/piecewise comparison is likewise a structural stress test, not an estimate of real response curvature.

7.3 Parameter Sources and Model Calibration

The numerical demonstration combines problem facts, methodological references, fitted synthetic parameters, and management assumptions. These sources support different kinds of claims. We identify them explicitly so that an enterprise pilot can replace assumed values without treating them as observed company data.

The first pilot should prioritize reliable failure denominators and recovery evidence. Next, obtain varied action histories to separate onset delay from persistence. Only then calibrate money-valued response curves for procurement. Each parameter record stores its unit, source field, effective date, method, uncertainty range, approver, and version. No expert panel was conducted in this work.

Table 14: Sources and what each can legitimately support.

Evidence class	Content used here	Required before deployment
Problem facts	Port workflow; confidentiality; P/T/R requirements	Confirm local scope and deadlines
Public references	Quality/provenance concepts; filtering; scoring; robustness	Check applicability and version
Scenario inputs	Costs, staffing, response curves, loss, generator coefficients	Replace with quotes, pilots, finance records
Fitted synthetic parameters	LCF persistence and delays; CRM coefficients	Refit on internal histories and independent outcomes
Management decisions	Reference mix, recovery floor, risk envelope	Document owner approval and rationale

8 Implementation of the Assessment Framework

8.1 Staffing and Training

A total headcount can conceal an uncovered night shift or a missing specialist skill. We assess staff availability by role, skill, and shift for data engineering, analytics, stewardship, interface operations, and incident response. P2 measures coverage of required cells; P1 and delayed P3 tests check practical competence. For skill r and shift h , the workload gap gives

$$\text{FTEgap}_{rh} = \max \left\{ 0, \frac{\text{required task hours}_{rh} - \text{qualified available hours}_{rh}}{\text{productive hours per FTE}_h} \right\}. \quad (34)$$

Round and schedule only after accounting for simultaneous shift requirements. A permanent, recurring gap supports hiring; a temporary specialist need supports contracting; an adjacent skill gap supports training when release time and retention are credible. Mixed staffing is often feasible, but its cost, lead time, turnover, and knowledge-transfer obligations belong in the project menu. No defensible number of hires can be calculated from the problem's withheld staffing data.

8.2 Technology Selection and Pilot Testing

Software specifications and vendor demonstrations do not necessarily predict performance in a port handoff. We compare candidate architectures through the same business acceptance tests: correct logical delivery after retries, recovery within the approved target, interoperable identities, controlled access, traceable records, exportable meta-data, and latency at peak workload. T1–T4 measure the service outcomes of those tests.

Table 15: Architecture comparisons the same model can support.

Question	Single suite	Multiple products
Integration effort	Test internal semantic consistency	Test cross-vendor interfaces and identities
Common failure	Model shared outage concentration	Model shared network/authentication dependencies
Exit and migration	Test full export and restore	Test data portability across components
Economics	License, migration, operations, training	Same costs plus interface maintenance

Start a controlled pilot on one handoff, compare against the current service using the same workload mix, and preserve a rollback path. A vendor's claimed accuracy is not interchangeable with an independently audited C2 rate. Only a verified improvement and a feasible resource plan justify wider rollout.

8.3 Data Governance and Routine Monitoring

When data rules change without a responsible owner, errors can recur even after a technical fix. We assign each critical dataset a business owner and technical steward, with a documented purpose, schema, quality rules, retention conditions, and access path. The owner decides business risk, the steward implements controls, and an

independent reviewer checks the evidence. Routine monitoring connects those responsibilities to incidents and corrective action.

Table 16: Monitoring and response protocol. Thresholds come from approved service commitments.

Frequency	Measure	Owner	Response
Daily	Availability, late critical handoffs, missing logs, security events	Operations and security	Incident triage; preserve evidence; notify affected users
Monthly	KPIs, missing bounds, skills, issue closure, C2	Data owners and HR	Publish versioned report and corrective actions
Quarterly	Calibration, capability drift, portfolio delivery	Model owner and finance	Revalidate; update constraints and budgets
Annual	Indicator scope, access, retention, recovery	Governance committee	Independent review and redesign where needed

An incident, a new data schema, a changed contract, a system migration, or calibration deterioration triggers an extra review. Do not wait for the annual cycle. An illustrative review rule is two successive monthly calibration errors exceeding an internally approved tolerance; the tolerance must be calibrated for sample size rather than chosen to make alerts disappear.

8.4 Implementation Schedule

The existing experiment uses 48 months of history, whereas a new assessment starts with inconsistent definitions and limited records. We separate the first 90 days of measurement setup from the later fitting of dynamic models. Each stage establishes the evidence needed for the next:

1. **Days 1–30: establish scope.** Inventory the two chains, critical fields, owners, and service commitments. Create expected-event registries and access permissions.
2. **Days 31–60: run a measurement pilot.** Compare source documents with logs, quantify unknowns, test recovery, and identify uncovered skill-slots.
3. **Days 61–90: approve provisional decisions.** Publish an ungraded baseline with uncertainty; gather vendor quotes and staffing costs; select a reversible pilot portfolio.
4. **After sufficient history: fit dynamic models.** The 48-month experiment does not imply a 90-day pilot can estimate all delays. Use direct KPI bounds until enough varied observations accumulate.

The customer dashboard shows current incident status, measured handoff reliability, uncertainty, reporting coverage, unresolved issues, and the date of independent review. Commercially sensitive personnel and financial fields remain internal.

9 Model Extension

9.1 Application to Ports of Different Sizes

A small port should not be penalized for lacking technology it does not need. Retain the people–technology–process structure, denominator rules, and chain-success definition, but recalibrate required service periods, staffing matrices, deadlines, workload controls, and costs. A small port may collect monthly audit samples and use a simple catalog; a large port may need continuous event feeds, segmented ownership, and shared-platform risk analysis. Both must demonstrate the same underlying ability to deliver agreed information.

Raw success rates can differ solely because one port handles more complex cargo or night-shift work. A common reference composition removes that part of the difference. For cargo/shift strata h , standardize each port b by

$$p_b^{\text{std}} = \sum_h \pi_h^{\text{ref}} p_{bh}, \quad \sum_h \pi_h^{\text{ref}} = 1. \quad (35)$$

For example, two equally capable ports have 0.95 success on routine tasks and 0.75 on complex tasks. A 90/10 task mix gives 0.93, whereas a 50/50 mix gives 0.85. At a common 70/30 reference mix, both score 0.89. The

eight-point raw difference is workload composition, not evidence of a capability gap. This arithmetic illustration shows why standardization is needed; it does not estimate any real port.

9.2 Application to Trucking Companies

The port handoff continues into dispatch and delivery, where mismatched identifiers or status definitions can recreate the same information failures. Trucking companies can use compatible event definitions, while replacing port-specific stages and deadlines with their own. The correspondence is shown below.

Table 17: Mapping the port framework to trucking.

Port element	Trucking counterpart	Common test
Arrival and inland schedule	Order, dispatch, appointment	Timely consistent identifiers
Yard location and release	Vehicle location and pickup authority	Current location and authorized release
Vessel loading confirmation	Proof of delivery	Complete, verifiable closing record
Port data roles	Dispatcher, driver, analyst, steward	Role-specific practical competence

The truck chain is order receipt → dispatch → location/ETA update → pickup or delivery → confirmed receipt. Refit its response coefficients and deadlines; do not copy the port's 0.6/0.4 weights or score thresholds. Sharing the metric benefits ICM through better arrival planning, fewer conflicting status records, faster attribution of data errors, and a common basis for service reviews.

At the port-carrier boundary, both parties confirm the same handoff identifier and timestamps. Joint success must be measured from matched records: multiplying separately reported port and carrier success rates assumes independence and can conceal common platform failures. Each party retains its own sensitive records while sharing authorized exception summaries.

10 Letter to ICM Corporation's customers

To: ICM Corporation's port users

From: The D&A assessment consulting team

Subject: Making your cargo information more reliable and accountable

Date: September 19, 2026; retrospective study of the 2022 case

Dear Port Users,

You depend on accurate arrival information, customs status, container locations, and pickup schedules. A delay or contradiction in one of these records can disrupt your own operations. We propose a measurement and improvement program that makes these failures visible and gives ICM a clear process for correcting them.

The program assesses whether employees have the skills needed for their jobs, whether technology delivers information when required, and whether data have accountable owners and controlled access. It also checks the complete handoff, from the first notice to the final confirmed transfer. A system will not receive credit simply because it has purchased new software.

Each expected handoff will be counted, including those with missing or late records. Independent checks will compare selected records with their underlying evidence. Reports will distinguish verified results from unknown outcomes, show current service incidents, and explain when a measurement has been revised. This allows you to understand both performance and the limits of the available evidence.

Your information will be used only for approved purposes. Access responsibilities, data retention, and authorized sharing will be documented. Confidential employee details and commercial records will remain within the appropriate organization. Public or customer reports will use approved summaries while preserving an internal audit trail.

When the evidence reveals a gap, ICM can compare training, staffing, governance, integration, and recovery improvements under its actual budget and operating constraints. Proposed changes will be tested on a limited scale, with clear responsibilities and a rollback path. We recommend checking current service conditions daily, reviewing performance monthly, and reassessing the improvement plan quarterly.

Carriers can participate through agreed shipment identifiers, compatible status definitions, and jointly confirmed timestamps. These shared measures should help the port and its customers identify where a handoff failed and resolve discrepancies without exchanging unnecessary confidential records.

This study has not inspected ICM's confidential systems. Its numerical examples are simulations, so we cannot certify the company's present maturity or promise a particular improvement. The proposed approach earns confidence through evidence: disclose what has been measured, acknowledge what remains uncertain, and track whether corrective actions work.

We recommend that ICM publish an initial measured baseline after an independently reviewed pilot and provide customers with a clear route for reporting inaccurate or delayed information. Your feedback should become part of the same tracked correction process.

Sincerely,

The D&A assessment consulting team

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References support methods and concepts, not the numerical scenario assumptions. The NIST citation uses its 2018 version for the 2022 case. Visual references for Figures 1–7 are [9, 10, 11, 12, 13, 14, 15], respectively; all plotted values are from this study. Semantic icons in Figures 1 and 2 use adapted Tabler components[16]; sources and license notices accompany the figure assets.

11 Reproducibility and AI-use disclosure

The archive includes complete code, raw synthetic records, fitted parameters, tables, and vector figures. Run `python code/run_models.py` and `python code/export_tables.py`, then compile both LaTeX entry points using the supplied build script. Random seed 2022 and dependency versions are recorded; `code/verify_results.py` checks numerical consistency. The generating and fitted parameters remain distinct in code and saved results.

OpenAI Codex assisted with model analysis, source lookup, code, bilingual drafting, and layout. Numerical claims come from executed experiments, including unsuccessful checks. The user supplied the problem, initial framework, layout example, and revised writing requirements. No corporate system access, expert interviews, certification, or independent peer review occurred. The study was prepared on September 19, 2026, and this disclosure counts toward the page limit.

12 Strengths and Limitations

The framework has four practical strengths. First, requirement coverage links every selected indicator to a business need, while mandatory governance and recovery checks remain protected. Second, the three core models form a traceable chain from audit records to capability states, service probabilities, and feasible investments. Third, explicit delays, diminishing returns, and quarterly constraints make the recommended menu operationally interpretable. Fourth, saved code, source counts, comparison models, and sensitivity results make both versions reproducible. Two limitations remain: the synthetic scenarios and finite project menu cannot fully represent the diversity of real ports; and the current uncertainty estimates require further calibration, as shown by 66.7% state-interval coverage and 5% budget violations outside the planning envelope. The next deployment step is therefore an independently audited enterprise pilot, using real event records and cost data to recalibrate the same framework.